Centre for UG and PG Studies (CUPGS), BPUT, Odisha

Department of Electrical Engineering

Power System -II (EEPC3004)

6th Sem B Tech & Dual Degree (B. Tech & M. Tech)- Electrical Engineering - 2025-26

Assignment Sheet-2

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Short type Questions:

1. Differentiate between economic load dispatch and unit commitment problem.
2. Differentiate between Hot Reserve and Cold Reserve.
3. What are the coordination equations with reference to economic load dispatch of thermal plants?
4. Write the Lagrange function for the thermal plants considering the loss function.
5. What is spinning reserve?
6. What are the two basic generator control loops? Briefly explain.
7. Draw the block diagram of a generator and briefly explain.
8. Draw the block diagram of a generator with load modelling and briefly explain.
9. Draw the block diagram of a prime mover and briefly explain. Comment on its time constant.
10. Draw the block diagram of a governor for LFC and briefly explain. Comment on its time constant.
11. Two turboalternators rated for 110 MW and 210 MW have governor drop characteristics of 5 per cent from no load to full load. They are connected in parallel to share a load of 250 MW. Determine the load shared by each machine assuming free governor action.
12. What are Penalty factor and its significance?
13. What is the purpose of adding a secondary integral loop of control in load frequency control loop?
14. What do you mean by Area Control Error (ACE)?
15. Briefly explain the significance of ACE.
16.

Long Type

1. Derive the solution of the economic load dispatch problem of a two generator system considering the transmission losses.
2. The input/output curve of 3 thermal plants in MBtu/h are given as: $H_1=510+7.2P_1+0.00142P_1^2$, $150<= P_1<=600$, fuel cost = 1.1\$/MBtu, $H_2=310+7.85P_2+0.00194P_2^2$, $100<= P_2<=400$, fuel cost = 1.0\$/MBtu, $H_3=78+7.97P_3+0.00482P_3^2$, $50<= P_3<=200$, fuel cost = 1.0\$/MBtu, P_1, P_2, P_3 are in MW. i) Find optimal load dispatch to supply a total load of 850. ii) modify the load dispatch when the fuel cost of unit 1 reduces to 0.9\$/MBtu.

3. The fuel cost of 3 thermal plants in Rs/h are given as: $C_1=500+5.3P_1+0.004P_1^2$, $C_2=400+5.5P_2+0.006P_2^2$ and $C_3=200+5.8P_3+0.009P_3^2$, P_1, P_2, P_3 are in MW. The total Load P_D is 800 MW. Neglecting losses and generation limits, find the optimal dispatch and the total cost in Rs/h.
4. The input output heat curves of three thermal plants are given as $H_1=510+7.2P_1+0.00142P_1^2$MBtu/h,$150<=P_1<=600$,$H_2=310+7.85P_2+0.00194P_2^2$MBtu/h,$100<=P_2<=400$, $H_3=78+7.97P_3+0.00482P_3^2$, $50<= P_3<=200$, P_1, P_2, P_3 are in MW. Fuel $cost_1$=1.1\$/MBtu, Fuel $cost_2$=1.0\$/MBtu, Fuel $cost_3$=1.2\$/MBtu. Construct a priority list based unit commitment table for the above units.
5. The incremental fuel rate curves of three thermal plants are given as $7000P_1+3.0P_1$ k-cal/MW-hr,$50<=P_1<=500$, $7500P_2+2P_2^2$ k-cal/MW-hr, $40<=P_2<=400$, $8000P_3+4P_3^2$ k-cal/MW-hr, $20<= P_3<=200$, P_1, P_2, P_3 are in MW. Fuel $cost_1$=1.1Rs/kcal, Fuel $cost_2$=1.05Rs/kcal, Fuel $cost_3$=1.2Rs/kcal. Construct a priority list based unit commitment table for the above units.
6. What is Unit Commitment problem? State the various methods to solve this.
7. Draw the schematic diagram of load frequency and excitation voltage regulators of a turbo-generator.
8. Draw and explain the block diagram of a two-area system LFC with only primary loop.
9. Draw the schematic diagram of a load frequency control loop and explain each of the blocks.
10. Two generators rated 200 MW and 400MW are operating in parallel. The droop characteristics of their governors are 4% and 5% respectively from no load to full load. Assuming that the generators are operating at 50Hz at no load, how would a load of 600MW be shared between them? What will be the system frequency then? Assume free governor operation.
11. Two generators rated 200MW and 400 MW are operating in parallel. The droop characteristics of their governors are 4% and 5% respectively from no load to full load. Assuming that the generators are operating at 50Hz at no load, how would a load of 600MW be shared between them? What will be the system frequency at this load? Assume free governor operation. Repeat the problem if both governors have a droop of 4%.
12. Two turboalternators rated for 110 MW and 210 MW have governor drop characteristics of 5 per cent from no load to full load. They are connected in parallel to share a load of 250 MW. Determine the load shared by each machine assuming free governor action.
13. An isolated power station has following parameters, Turbine time constant TT =0.5 sec, Governor time constant Tg=0.2 sec, generator inertia constant H= 5sec, and Governor speed regulation=R pu. The load varies by 0.8 percent for a 1 percent change in frequency i.e. D=0.8. Use (a) Routh-Hurwitz array to find out to find the range of R for control system stability. (b) the governor regulation R is set to 0.05 per unit. The

turbine rated output is 250 MW at nominal frequency of 60 Hz. A sudden change of load of 50 MW occurs Find the steady state frequency deviation in Hz.